Advanced Micro Devices

Advanced Media Framework – HQ Scaler

Programming Guide

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1 Introduction

AMF HQ Scaler is a technique for achieving high-end video upscaling results from lower resolution video inputs. This document provides a complete description of the AMD Advanced Media Framework (AMF) Video HQ Scaler Component. This component performs the following functions:

- HQ Scaling
- Sharpening

2 AMF Video HQ Scaler Component

Video HQ scaler accepts input frames stored in `AMFSurface` objects wrapping DirectX 11/12 textures, Vulkan surfaces, OpenCL surfaces. The output is placed in `AMFSurface` objects wrapping DirectX 11/12 textures, OpenCL surfaces, Vulkan surfaces, depending on the component configuration.

Include `public/include/components/HQScaler.h`

2.1 Component Initialization

The AMF Video HQ Scaler component should be initialized using the following sequence:

1. Create an AMF Context and initialize it for one of the following:
    i. DirectX 11
    ii. DirectX 12
    iii. Vulkan
    iv. OpenCL
2. Configure the HQ Scaler component by setting the necessary properties using the `AMFPropertyStorage::SetProperty` method on the HQ Scaler object.
3. Call the `VideoHQScaler::Init` method of the video HQ Scaler object.

2.2 Configuring the HQ Scaler

The HQ scaler supports the following input and output formats:

1. BRGA
2. NV12
3. RGBA
4. R10G10B10A2
5. RGBA_F16
6. P010

The output format must be same as the input and the format conversion is not supported. The parameters of the output stream are set using the following properties:

Name (prefix "AMF_HQ_SCALER_")	Type
ENGINE_TYPE	AMF_MEMORY_TYPE
OUTPUT_SIZE	AMFSize
KEEP_ASPECT_RATIO	Bool
FILL	Bool
FILL_COLOR	AMFColor
ALGORITHM	amf_int64
FROM_SRGB	Bool
SHARPNESS	Float

Table 1. AMF HQ Scaler properties of the output stream

Name: `AMF_HQ_SCALER_ENGINE_TYPE`

Values: `AMF_MEMORY_DX11` , `AMF_MEMORY_DX12` , `AMF_MEMORY_VULKAN` , `AMF_MEMORY_OPENCL`

Default Value: `AMF_MEMORY_DX11`

Description: Specifies the memory type of output surfaces. Surfaces are allocated internally by the HQ Scaler component.

Name: `AMF_HQ_SCALER_OUTPUT_SIZE`

Values: A valid size.

Default Value: `N\A`

Description: Output image resolution specified as `AMFSize` . Scaling will be performed when this property is set.

Name: `AMF_HQ_SCALER_KEEP_ASPECT_RATIO`

Values: `true` , `false`

Default Value: `false`

Description: Force the scaler to keep the aspect ratio of the input image when the output size specified by the `AMF_HQ_SCALER_OUTPUT_SIZE` property has a different aspect ratio.

Name: `AMF_HQ_SCALER_FILL`

Values: `true` , `false`

Default Value: `false`

Description: Specifies whether the output image outside the region of interest, which does not fill the entire output surface should be filled with a solid color. The fill color is specified using the `AMF_HQ_SCALER_FILL_COLOR` property.

Name: `AMF_HQ_SCALER_FILL_COLOR`

Values: `(0,0,0,0)` ... `(255,255,255,255)`

Default Value: `(0,0,0,255)`

Description: Fill color specified as `AMFColor` to fill the area outside the output rectangle. Applicable only when the `AMF_HQ_SCALER_FILL` property is set to `true`.

Name: `AMF_HQ_SCALER_ALGORITHM`

Values:

Name	Description
`AMF_HQ_SCALER_ALGORITHM_BILINEAR`	Bilinear scaling algorithm.
`AMF_HQ_SCALER_ALGORITHM_BICUBIC`	Bicubic scaling algorithm.
`AMF_HQ_SCALER_ALGORITHM_POINT`	Point (nearest-neighbor) scaling algorithm.
`AMF_HQ_SCALER_ALGORITHM_VIDEOSR1_0`	VideoSR1.0 scaling algorithm. This algorithm is based on FSR 1.0.
`AMF_HQ_SCALER_ALGORITHM_VIDEOSR1_1`	VideoSR1.1 scaling algorithm. This algorithm is intended for specific internal integrations and is exposed purely for experimental use. VideoSR1.1 is only supported when `AMF_HQ_SCALER_ENGINE_TYPE` is set to `AMF_MEMORY_DX11` or `AMF_MEMORY_DX12` and the input and output formats are not NV12 or P010.

Default Value: `AMF_HQ_SCALER_ALGORITHM_VIDEOSR1_0`

Description: Specifies scaling method.

Name: `AMF_HQ_SCALER_FROM_SRGB`

Values: `true`, `false`

Default Value: `true`

Description: Convert color space from linear to SRGB.

Name: `AMF_HQ_SCALER_SHARPNESS`

Values: Float in the range of `[0.0, 2.0]`

Default Value: `0.5`

Description: Control VideoSR scaler sharpening. Applicable only when the `AMF_HQ_SCALER_ALGORITHM` property is set to `AMF_HQ_SCALER_ALGORITHM_VIDEOSR1_0` or `AMF_HQ_SCALER_ALGORITHM_VIDEOSR1_1`.

2.3 Submitting Input and Retrieving Output

Once the HQ Scaler component is successfully initialized, you may start submitting input samples to it. Input samples must be submitted as `AMFSurface` objects.

At the same time poll for output by calling `AMFComponent::QueryOutput` on the HQ Scaler object. Polling for output samples can be done either from the same thread or from another thread.

Suspend submission of input samples briefly when `AMFComponent::SubmitInput` returns `AMF_INPUT_FULL`. Continue to poll for output samples and process them as they become available.

2.4 Terminating the HQ Scaler Component

To terminate the HQ Scaler component, call the `Terminate` method, or simply destroy the object. Ensure that the context used to create the HQ Scaler component still exists during termination.